4.7 The Calculus of Variations

WORKSHEET

The question

STEP III 2014, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q4

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Why is it not obvious that $I \geq 0$? Try $y = 1 - x$, which satisfies $y(1) = 0$.

STAGE 1 Completing the square

Ask yourself The integrand $y^2 \sec^2 x + 2yy' \tan x$ is an exact derivative. Of what?

STAGE 2 When is $I = 0$?

Ask yourself If the integral of a continuous non-negative function is zero, what can you say about the function? So what equation must y satisfy if $I = 0$?

STAGE 3 The general case

Ask yourself Repeat Stage 1 with $\int_0^1 (y' + ay\tan bx)^2\,\mathrm{d}x$. What must b be for the leftover terms to form an exact derivative?

Pause and try Where can this argument break down? (Look at $\tan ax$ on the interval $0 \le x \le 1$.)

STAGE 4 The critical case

Ask yourself When $a = \frac{\pi}{2}$, the proof fails. In Stage 2, the function making $I = 0$ would have satisfied $y' + y\tan x = 0$. What is the analogous equation now, and what are its solutions?

What's really going on

The brachistochrone and the birth of a subject

In 1696 Johann Bernoulli challenged the mathematicians of Europe: find the curve down which a bead slides from one point to another in the least time. Solutions came from Newton (anonymously, overnight; Bernoulli said he recognised "the lion by his claw"), Leibniz, l'Hôpital and Jakob Bernoulli. The answer is a cycloid. In the 1750s Euler and the young Joseph-Louis Lagrange turned these special tricks into a general method: if y makes $\int F(x, y, y')\,\mathrm{d}x$ stationary, then y satisfies the *Euler–Lagrange equation*

$$\frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial F}{\partial y'} = \frac{\partial F}{\partial y}.$$

For our J, $F = (y')^2 - a^2y^2$, and the Euler–Lagrange equation is $y'' + a^2y = 0$: simple harmonic motion.

Why $\pi/2$?

For $a < \frac{\pi}{2}$, J has a minimum value of 0, achieved only by $y \equiv 0$. At $a = \frac{\pi}{2}$ the minimum is still 0 but a second function achieves it. For $a > \frac{\pi}{2}$, the same function $y = \cos\frac{\pi x}{2}$ gives

$$J = \frac{\pi^2}{8} - \frac{a^2}{2} < 0,$$

and by scaling y up, J can be made as negative as you like: there is no minimum at all (Figure 4.7.1). The critical value is the first a for which $y'' + a^2y = 0$ has a non-zero solution satisfying the boundary conditions ($y(1) = 0$, and a "natural" condition $y'(0) = 0$ at the free end). That is an *eigenvalue problem*, and $a^2 = \frac{\pi^2}{4}$ is its smallest eigenvalue. Compare Section 3.7, where pinning the function at both ends of an interval of length π gave critical value 1; here the interval has length 1 and only one end is pinned, which is like half of an interval of length 2, so the critical a is $\frac{\pi}{2}$.

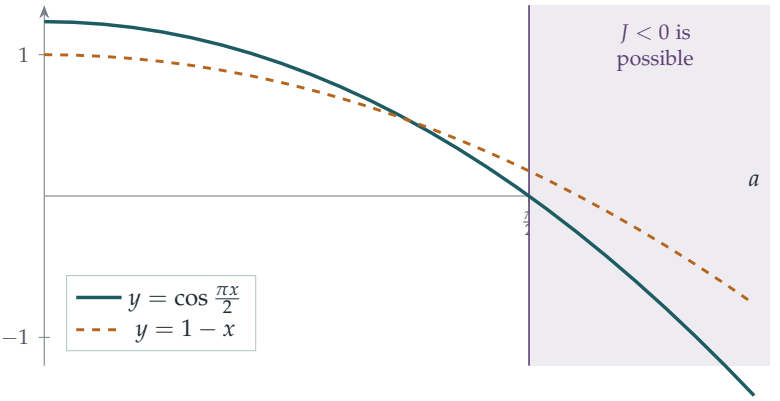


Figure 4.7.1 The value of J against a for two test functions with $y(1) = 0$. For $y = 1 - x$, $J = 1 - \frac{a^2}{3}$ becomes negative only when $a > \sqrt{3}$. For $y = \cos \frac{\pi x}{2}$, $J = \frac{\pi^2}{8} - \frac{a^2}{2}$ becomes negative as soon as $a > \frac{\pi}{2}$, and no function does better: for $a < \frac{\pi}{2}$, $J \geq 0$ for every y.

Physics and the principle of least action

The calculus of variations turned out to describe nature itself. In the eighteenth and nineteenth centuries it emerged that the laws of motion can be stated as: a mechanical system moves so as to make an integral called the *action* stationary. Light, similarly, travels along the path of least time (Fermat's principle), which explains refraction. In the twentieth century the same idea became the foundation of quantum mechanics, through Richard Feynman's "sum over paths". The quadratic integrals in this question are the simplest examples, and the eigenvalue problem they lead to reappears as the energy levels of a quantum oscillator in Section 8.6.

Try this next

1 **Beyond the critical value.** Show that if $a > \frac{\pi}{2}$ then $y = \cos \frac{\pi x}{2}$ gives $J < 0$.

2 **The Euler–Lagrange equation.** HARDERSuppose y minimises J among functions with $y(1) = 0$. For any function η with $\eta(1) = 0$ and any number ε, $J(y + \varepsilon\eta) \geq J(y)$. Expand $J(y + \varepsilon\eta)$ in powers of ε and deduce that $\int_0^1 (y'\eta' - a^2 y\eta)\,\mathrm{d}x = 0$. Integrate by parts to show that $y'' + a^2 y = 0$ and $y'(0) = 0$ (assuming y is smooth enough).

3 **A longer interval.** Show by a substitution that if $y(L) = 0$, then $\int_0^L \left((y')^2 - a^2 y^2\right)\mathrm{d}x \geq 0$ for all y exactly when $a \leq \frac{\pi}{2L}$, and that a non-zero y gives equality only when $a = \frac{\pi}{2L}$.

4 **The cycloid.** Look up the brachistochrone problem and find the parametric equations of the cycloid. Show that a bead released from rest anywhere on an upside-down cycloid reaches the bottom in the same time (the *tautochrone* property, found by Huygens in 1659).

Further reading

- **Paul Nahin, *When Least Is Best*** (Princeton University Press). The history and mathematics of optimisation, including the brachistochrone, written for students.
- **Richard Feynman, *The Feynman Lectures on Physics*,** Volume II, Chapter 19 (“The principle of least action”). Free online.
- **Bruce van Brunt, *The Calculus of Variations*** (Springer). An undergraduate textbook; the first chapters need only A-level calculus.

The full guided solution is in Section 4.7:
https://beyondstep.keithmcnulty.org/ch04/variations.html